



American International University-Bangladesh (AIUB)

Department of Computer Science Faculty of Science & Technology (FST)

Project Title:

Real-Time Tracking and Automated Fare Management System

A Software Engineering Project Submitted

By

GROUP:08

Semester: Fall_24_25		Section:	Group Number:	
SN	Student Name	Student ID	Contribution (CO3+CO4)	Individual Marks
01	MASUDUR RAHMAN	23-50797-1	20	
02	SAIF AHMED AYON	22-47572-2	20	
03	SAFAYAT HASAN	22-46676-1	20	
04	MD.ARIF BILLAH	23-50470-1	20	
05	SAMIN, K.M.M	22-49904-3	20	

The project will be Evaluated for the following Course Outcomes

CO3: <i>Select</i> appropriate software engineering models, project management roles and their associated skills for the complex software engineering project and evaluate the sustainability of developed software, taking into consideration the societal and environmental aspects	Total Marks	
	Appropriate Process Model Selection and Argumentation with Evidence	[5 Marks]
	Evidence of Argumentation regarding process model selection	[5Marks]
	Analysis the impact of societal, health, safety, legal and cultural issues	[5Marks]
CO4: <i>Develop</i> project management plan to manage software engineering projects following the principles of engineering management and economic decision process	Submission, Defense, Completeness, Spelling, grammar and Organization of the Project report	[5Marks]
	Total Marks	
Develop the project plan, its components of the proposed software products		[5Marks]

Identify all the activities/tasks related to project management and categorize them within the WBS structure. Perform detailed effort estimation correspond with the WBS and schedule the activities with resources	[5Marks]	
Identify all the potential risks in your project and prioritize them to overcome these risk factors.	[5Marks]	

Description of Student's Contribution in the Project work

Student Name: MASUDUR RAHMAN Student ID: 23-50797-1 Contribution in Percentage (%): 20% <u>Contribution in the Project:</u> ▪ Contribution Description 1 ▪ Contribution Description 2 MASUDUR RAHMAN Signature of the Student
Student Name: SAMIN, K.M.M Student ID: 22-49904-3 Contribution in Percentage (%): 20% <u>Contribution in the Project:</u> ▪ Contribution Description 1 ▪ Contribution Description 2 SAMIN, K.M.M Signature of the Student
Student Name: SAFAYAT HASAN Student ID: 22-46676-1 Contribution in Percentage (%): 20% <u>Contribution in the Project:</u> ▪ Contribution Description 1 ▪ Contribution Description 2 SAFAYAT HASAN Signature of the Student
Student Name: SAIF AHMED AYON Student ID: 22-47572-2 Contribution in Percentage (%): 20% <u>Contribution in the Project:</u> ▪ Contribution Description 1

▪ Contribution Description 2

SAIF AHMED AYON

Signature of the Student

Student Name: **MD. ARIF BILLAH**

Student ID: 23-50470-1

Contribution in Percentage (%): 20%

Contribution in the Project:

- Contribution Description 1
- Contribution Description 2

MD.ARIF BILLAH

Signature of the Student

1. PROJECT PROPOSAL

1.1 Background to the Problem

- Write the background description that helps putting your project into the right context of a problem domain and gives everyone involved a common view of the project.
- What is the root cause of this problem? Why this problem is so important to consider?

1.2 Solution to the Problem

- Describe what is your project/thesis objective? What solutions are you going to provide to solve the above-mentioned problems?
- What are the solutions you are going to propose to deal with the problem? Why is this solution is particularly appropriate to solve the problem? Is the solution feasible to meet the business objective?
- Describe the basic functionalities of your proposed solution that makes the best use of state-of-art technology and produces a significant result that is likely to have a major impact on societal, health, safety, legal and cultural issues. Provide a deep insight that demonstrates and preset a creative solution to the real-life problem.
- Describe the target group of users of your solution? And how they will benefit from your proposed solution to the problem?
- Describe the contribution of your project to the development of scientific results that are identified and well documented.
- Provide a literature review on what are the other studies that have discussed the same topic of yours in the literature and explain how your study has utilized and extended the problems of existing studies.
- Provide a description of all the existing studies presented in the problem area. What are the

- existing software solutions (for project) are available to solve the aforementioned problems?
- What are the existing software solutions available to solve the aforementioned problem? And how your proposed solution is going to extend them in providing more benefits to the users?

2. SOFTWARE DEVELOPMENT LIFE CYCLE

2.1 Process Model

- Provide an analysis regarding the nature and environment of the software that you are going to develop and select the most suitable method(s) to develop the software.
- Present your arguments based on your analysis about why your selected method(s) is the best choice among all other methods to develop your proposed software.
- Presents sufficient amount of evidence to support argument for your model selection in developing your proposed solution.

2.2 Project Role Identification and Responsibilities

- Identify all the roles/stakeholders in the software/project management activities in software development.
- Describes the responsibilities of the role in the software development.

Text Format:

- Style: Times New Roman
- Size: 12
- Space: 1.0
- Alignment: Justify
- Length: Maximum 6 pages (including cover page)

Rubric for Project Assessment (CO3)

Criteria	Marks distribution (Max 3X5= 15)				Acquired Marks
	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	
Selection of Software Engineering Models	Does not articulate a position or argument of choosing appropriate model. Does not present any evidence to support the arguments for the choice of the model	Articulates a position or argument for choosing models that is unfocused or ambiguous. Presents incomplete/vague evidence to support argument for model choice	Articulates a position or argument of choosing models that is limited in scope. Does not present enough evidence to support the argument for the choice of the model	Clearly articulates a position or argument for the choosing software engineering models. Presents sufficient amount of evidence to support argument for the model selection	
Role identification and Responsibility Allocation	The project has poor project management plans for identifying roles and assigning the responsibilities	Identify few roles in the project management where some of the roles are left alone with any project responsibilities	Identify most of the roles in the project management and assign their responsibilities	Well planned project with proper role identification and responsibility allocation in the project management activities	
Impact identification					
Formatting and Submission	Project report is not complete and Several errors in spelling and grammar. Present a Confusing	Some errors in spelling and grammar. Some problems	Few errors in spelling and grammar. Presents most of the details in	Project report is complete and No errors in spelling and grammar. Consistently	

	organization of concepts, supporting arguments, and real-life example. Sentences rambling, and details are repeated.	of organizing the answer in a logical order of defining, elaborating, and providing real-life examples.	a logical flow of organization in definition, details, and example.	presents a logical and effective organization of definition, details, and real-life example of the topic.	
Acquired marks:					
CO Pass / Fail:					

Rubric for Project Assessment (CO4)

Marks Distribution (Maximum 3X5=15)					
Marking Criteria	Inadequate (1-2)	Satisfactory (3)	Good (4)	Excellent (5)	Acquired Marks
Project Planning	No background information regarding the project is given; project goals and benefits are missing.	Insufficient background information is given; project goals and benefits are poorly stated	Sufficient background information is given; the purpose and goals of the project are explained.	Thorough and relevant background information is given; project goals are clear and easy to identify.	
Effort Estimation and Scheduling	Student vaguely discuss the impact of societal, health, safety, legal and cultural issues in their project	Student provided with partial relevance to the impact of societal, health, safety, legal and cultural issues in their project	Student fairly provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	Student comprehensively provided the analysis to the impact of societal, health, safety, legal and cultural issues in their project	
Risk Management	Ambiguous representative example.	Partially identify / indicate towards real-life example.	Real-life example is fairly connected towards the definition.	Comprehensively defend with real life example.	
Acquired Marks:					
CO Pass / Fail:					

CO5 [PO-i-2]: Perform as an effective team member or leader in diverse team settings and solve multi-disciplinary problems in computer science and engineering domain.				
Assessment Attribute/Criteria	Missing/ Incorrect (0)	Inadequate (1)	Satisfactory (2)	Excellent (3)
Taking responsibility	Does not perform assigned tasks; often misses meetings and, when present, does not have anything constructive to say; relies on others to do the work;	Partially performs all assigned tasks; attends meetings irregularly and occasionally participates and hence not reliable;	Performs all assigned tasks; attends meetings regularly and usually participates effectively. generally reliable;	Performs all tasks very effectively; attends all meetings and participates enthusiastically; very reliable.
Contributions	Never provides useful ideas when participating in a group discussion	Rarely provides useful ideas when participating in a group discussion	Sometimes provides useful ideas when participating in a group discussion	Routinely provides useful ideas when participating in a group discussion
Collaboration and Ability to Compromise	Not cooperative, unable to compromise and disrupts the team process.	Sometimes cooperative and rarely displays a positive attitude.	Usually cooperative, able to compromise and generally display positive attitude.	Always cooperative. Willingness to compromise. Always display positive attitude.
Valuing other team members (Working with others)	Often argues with teammates; doesn't let anyone else talk; occasional personal attacks and "put downs"; wants to have things done his way and does not listen to alternate approaches.	Seldom listens to others' points of view; occasionally behaves in an oppressive manner; tries to force their own ideologies on other.	Generally, listens to others' points of view; always uses appropriate and respectful language; tries to make a definite effort to understand others' ideas.	Always listens to others and their ideas; helps them develop their ideas while giving them full credit; always helps the team reach a fair decision.

1.1 Background Study:

Public transportation systems are vital for the smooth functioning of urban areas, providing an essential service to millions of passengers daily. Efficient bus services are crucial for reducing traffic congestion, lowering pollution levels, and providing affordable transportation options. However, traditional bus systems face challenges such as inaccurate schedules, inefficient route management, and a lack of real-time information for passengers. This study aims to explore the current landscape of bus tracking and fare calculation systems and the technologies that can be leveraged to improve these services. Commuters often experience uncertainty regarding bus schedules, leading to prolonged waiting times and missed connections. This unpredictability discourages potential users, pushing them towards private transportation options that exacerbate traffic congestion and environmental pollution. A root cause of this problem is the absence of real-time information on bus locations and arrival times. Without this information, both transit authorities and passengers struggle to plan and optimize travel routes effectively. Traditional fare collection methods, such as cash payments and paper tickets, are inefficient and prone to errors. They slow down boarding times, contribute to revenue leakages, and create inconvenience for both passengers and transit operators. The root cause here is the reliance on outdated technologies and manual processes, which cannot keep up with the demands of modern, fast-paced urban environments. Moreover, these systems often lack integration with other modes of transportation, hindering the creation of a seamless travel experience for commuters.

1.2 Solution to the Problem

Problems:

Root Cause:

- ✦ **Traffic Congestion:** Dhaka and other major cities in Bangladesh suffer from severe traffic congestion, causing buses to run late and making it difficult for passengers to rely on bus schedules.
- ✦ **Inefficient Fare Collection:** Traditional fare collection methods are prone to errors, fraud, and inefficiency, leading to revenue loss and inconvenience for passengers.
- ✦ **Lack of Real-Time Information:** Passengers often lack real-time information about bus locations and arrival times, making travel planning difficult and uncertain.
- ✦ **Unplanned Schedule:** Poor route planning and schedule management lead to longer travel times and underutilization of resources. □ Causes of Consideration.
- ✦ **Growing Urban Population:** As urban areas in Bangladesh grow, the demand for efficient public transportation increases, necessitating improvements in the bus system.

Technology Adoption: The increasing availability and affordability of GPS technology and smartphones make it feasible to implement advanced tracking and fare management systems.

- ✦ **Passenger Convenience:** By providing accurate real-time information and simplifying fare payments can significantly improve passenger satisfaction.

Project Objectives:

- ✦ The main objective of the project is to provide real-time tracking of bus services.
- ✦ Provide passengers with real-time information on bus locations, estimated arrival times, and route details through a mobile app.
- ✦ Implement automated fare calculation and payment systems to reduce errors, fraud, and inconvenience associated with manual fare collection.

Solution:

- ✦ Implementing real-time tracking of buses can help passengers to know the bus location.
- ✦ Implement a system that automatically calculates fares based on distance travelled or predefined fare rules.
- ✦ Develop user-friendly mobile apps and web portals where passengers can access real-time bus locations and schedules.
- ✦ Implementing tools for schedule management can reduce travel time.

Target Users:

- ✦ People
- ✦ Bus conductors and Drivers
- ✦ Bus companies
- ✦ Technology and Service providers

Basic Functionality:

- ✦ **Login:** The users can login to the system using their username and password. The login information provided on this page will be verified using the database.
- ✦ **Registration:** The system will ask for the user to give their full name, phone number, email, area, and car details. If all the information of the data is provided on the system, the page will redirect the user to another page for providing username and password for their account.
- ✦ **Tracking:** View the real-time location of buses on a map via a mobile app or web portal.

- ✦ **Route Information:** Access detailed information about bus routes, including all stops and transit times.

Search: Users can view the search criteria for finding a needed bus and route. They can be able to see the bus according to their destination and also able to know the correct fare.

- ✦ **Request submissions:** User can also find bus according to their comforts. Such as Ac and non-ac bus.
- ✦ **Book seat:** Customers can book seats.
- ✦ **Passenger Management:** Driver can see pick up location of the passenger.
- ✦ **Payment options:** The customer can pay the fair using the app through mobile banking.
- ✦ **Feedback:** It will offer users feedback.
- ✦ **Ratings and Complain:** Users can also give ratings and can complain to the authority through the software.

Contribution of project:

- ✦ The project will include advancements in transportation technology.
- ✦ The design of user-friendly application will help the user to find the correct bus easily.
- ✦ Improved understanding of passenger needs and preferences, informing future transport designs.
- ✦ Increases trust in the fair system and ensures passengers only pay for the distance they travel.
- ✦ Implementation of an electronic ticketing system will make the payment easier for passengers

Literature Review:

Research indicates that traditional bus systems rely on manual estimates of arrival times, which are often inaccurate due to factors like weather and traffic conditions. Implementing a wireless sensor network using GPS can provide real-time location information, enhancing the accuracy and reliability of bus services. Mobile applications can disseminate this data, offering users up-to-date information on bus positions, estimated arrival times, and recommended routes. This integration not only improves time management for users but also boosts overall service quality and satisfaction. Practical implementations, such as the Tran Santiago system in Chile, have demonstrated the effectiveness of combining smart card and GPS data for better policy and planning decisions. Furthermore, utilizing IoT allows for the real-time connection of various devices, facilitating the efficient collection and transmission of bus location data.

Existing Software:

The name of some existing software is given below:

Movita: Movita is a popular public transit app that provides real-time bus tracking, route planning, and fare information.

- ✦ Bus bud: Bus bud provides a platform for searching, comparing, and booking intercity bus tickets, including real-time tracking and fare management.

- ✦ Swiftly: Swiftly offers software solutions for public transit agencies, including real-time vehicle tracking, schedule adherence, and passenger information systems.

2.1 Process Model

Selection of Process Model:

Extreme Programming (XP) is one of the most widely used software development models. XP is a type of Agile methodology that emphasizes customer satisfaction, continuous improvement, and responsiveness to changing requirements. To apply XP, there are 12 practices to follow at an extreme level.

- ✦ Practice 1: The Planning - In XP, the planning game involves both developers and stakeholders to collaboratively prioritize features based on their value and feasibility.
- ✦ Practice 2: Small Releases - XP emphasizes delivering functional software in small, incremental releases.
- ✦ Practice 3: System Metaphor - The system metaphor is a simple, shared story that describes how the system works.
- ✦ Practice 4: Simple Design - XP advocates for the simplest design that works, avoiding unnecessary complexity.
- ✦ Practice 5: Continuous Testing - Continuous testing ensures that the software is always in a potentially shippable state.
- ✦ Practice 6: Refactoring - Refactoring involves restructuring existing code without changing its external behavior to improve readability and maintainability.
- ✦ Practice 7: Pair Programming - In pair programming, two developers work together at one workstation, with one writing code and the other reviewing it in real-time.
- ✦ Practice 8: Collective Code Ownership - Collective code ownership means that any team member can modify any part of the codebase.

Practice 9: Continuous Integration and Daily Build - Continuous integration (CI) involves regularly integrating code changes into a shared repository, followed by automated testing

- ✦ Practice 10: 40-Hour Work Week - XP advocates for sustainable work hours to maintain productivity and creativity.

Practice 11: On-Site Customer - An on-site customer (or a representative) is always available to provide feedback and clarify requirements

Practice 12: Coding Standards - Coding standards ensure that all code follows a consistent style and practices, making it more readable and maintainable

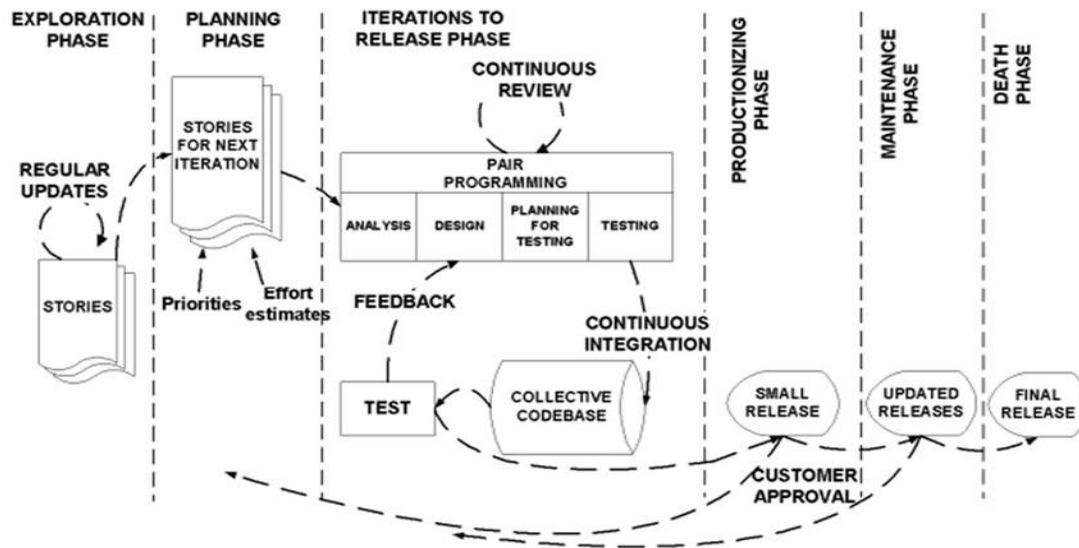


Fig-1: XP Process Model

Why was XP Process Model chosen for this project?

Extreme Programming (XP) is particularly suitable for this project due to its focus on flexibility, continuous feedback and high-quality code delivery. The nature of this project involves integrating real-time data for bus locations, dynamically displaying bus routes, and accurately calculating fares, all of which require robust and adaptive software solutions. XP's iterative and incremental development process allows the team to deliver functional software quickly, providing a working product that customers can interact with early in the project lifecycle. This early feedback is crucial for refining features based on real-world usage and user requirements, ensuring the system meets the actual needs of the users.

Moreover, the project environment demands rapid response to changing conditions, such as updates to bus routes or modifications in fare structures. XP's emphasis on flexibility and responsiveness is ideal in such a dynamic environment. Practices like Test-Driven Development (TDD) and Continuous Integration (CI) ensure that new code is continuously tested and integrated, maintaining system stability and reliability. Pair programming and collective code ownership

foster a collaborative environment where knowledge sharing and quick problem resolution are standard, reducing the risk of bottlenecks and enhancing code quality.

XP also promotes a high level of customer involvement through the on-site customer practice, ensuring that the development team has constant access to user feedback and can make necessary adjustments swiftly. This close collaboration helps in aligning the software closely with user expectations, leading to higher satisfaction and better usability. The rigorous adherence to coding standards and regular refactoring in XP further ensures that the codebase remains clean, maintainable, and scalable, which is essential for a project that will likely evolve and grow over time.

That's why XP Process Model is chosen for this project

The identity of roles or stakeholders in the software/project management activities in software development:

The responsibilities of the role in the software development:

- ✦ **Customer (On-Site Customer):** Provides detailed and clear requirements based on business needs. Prioritizes user stories and features to guide development. Offers continuous feedback on iterations to ensure the product meets business objectives.
- ✦ **Developer:** Writes and maintains the codebase following XP practices such as pair programming and test-driven development. Writes and runs unit tests to ensure code quality and functionality. Continuously improves and refactors the code to enhance performance and maintainability.
- ✦ **Designer:** Conducts research to understand user needs, behaviors, and pain points. Creates wireframes, mockups, and prototypes to visualize and test the user interface and experience. Conducts usability testing to gather feedback and iteratively improve the design. Works closely with programmers and customers to ensure the design is implemented effectively.
- ✦ **Tester:** Develops test plans and scenarios to ensure comprehensive coverage of user stories. Conducts exploratory and manual testing to identify issues and ensure quality. Writes automated tests to validate functionality and performance.
- ✦ **Tracker:** Monitors the progress of the development against the plan. Collects and analyzes metrics to track the team's performance and velocity. Identifies potential risks and issues and works with the team to mitigate them.
- ✦ **Manager:** Ensures that the team has the necessary resources to complete the project. Acts as a liaison between the team and external stakeholders. Provides support and removes obstacles that may hinder the team's progress. Fosters a positive and productive work environment aligned with XP values.

- ✦ **Coach:** Provides direction and guidance on XP practices and ensures the team adheres to them. Supports team members in understanding and implementing XP principles and practices. Helps the team resolve issues and improve their processes continuously.

Impact of the project in society:

Enhanced Commuter Experience

The Smart Bus project will provide real-time information on bus locations, estimated arrival times, and route recommendations. This will reduce waiting times, improve time management, and increase the overall convenience for commuters.

Increased Public Transport Usage

By offering a reliable and efficient public transportation system, more people will be encouraged to use buses instead of private vehicles. This can lead to reduced traffic congestion and lower pollution levels, contributing to a cleaner environment.

Improved Operational Efficiency

The integration of automated fare management will streamline fare collection, reduce human errors, and minimize fare evasion. This will lead to increased revenue for transportation authorities and better allocation of resources.

Economic Benefits

Efficient public transportation supports economic development by facilitating the mobility of the workforce and reducing transportation costs. Enhanced public transport systems can attract more businesses and tourism to the area, boosting local economies.

3.1 Project description

The proposed system aims to revolutionize public transportation by providing a user-friendly platform for bus tracking, seat reservation, and secure payment options. It is designed with the convenience of users in mind while ensuring robust functionality and security. The login feature allows users to access their accounts securely using a username and password. Login credentials are verified against the database, and successful login grants access to the user's home page. If incorrect credentials are entered, the system generates a verification code and sends it to the user's email for retrying. To prevent misuse, the system blocks accounts for one hour after three unsuccessful login attempts. This ensures a high level of security and protects user data. To use the system, users first need to register by providing necessary details such as **name, email, phone number, password, and area**. Registration requires a unique username and valid contact information, ensuring smooth onboarding. Once registered, users are directed to the login page to access their account. A major feature of the system is real-time bus tracking, which relies on **GPS data** to show bus locations on an interactive map. This feature allows users to search for buses by route or location, offering a practical and efficient way to plan their journeys. The reservation module lets users reserve seats on specific buses. They can **view fares, select their pickup points, and confirm their journey**. Once a reservation is made, users proceed to payment, which can be completed via popular mobile banking services like **Baksh, Nagad, or Rocket**. The system ensures that users with insufficient balances are notified, preventing failed bookings. After successful payment, **users receive a receipt via email**, and they are redirected to the homepage for further navigation. To ensure a seamless experience, the system is built to handle up to 10,000 concurrent users, maintaining a response time of less than **two seconds** for user queries. It guarantees 99.9% uptime and is designed to recover from failures within **five minutes**, ensuring high reliability. All communications are secured through HTTPS, and user data is encrypted both during transit and at rest, meeting stringent security standards. The system is also designed to be accessible and intuitive. Its interface is easy to use, with features accommodating users with disabilities. Mobile friendliness is a core consideration, ensuring smooth operation across various devices. The system integrates with third-party services like Google Maps for accurate tracking and mapping functionalities.

In terms of constraints, the system adheres to specific development technologies, such as JavaScript, React, and Node.js, and uses robust database management systems like MySQL or PostgreSQL. It complies with data privacy regulations like GDPR and accessibility standards like WCAG 2.1. Operationally, it depends on accurate GPS data and reliable internet connectivity for real-time updates. Deployment follows a defined schedule on a specified cloud platform with minimal downtime. Overall, the system prioritizes performance, reliability, security, and usability, providing an efficient and secure public transportation solution. By integrating advanced technologies and adhering to global standards, it ensures a seamless

and satisfying user experience for passengers, making public transport more accessible and convenient.

4.1 Functional Requirements

1. Software Login:

The software shall allow users to login with their given username and password. The login credentials (username and password) will be verified with database records. If the login successful, the home page of the user account will be displayed. If the username and/or password has been inserted wrong, the random verification code will be generated and sent to the user's email address by the system to retry login. If the number of login attempts exceed its limit (3 times), the system shall block the user account login for one hour.

Priority Level: High

Precondition: Users have valid user id and password

2. Registration:

The user will provide their **name, password, phone number, email** and **area in** the system. After registration, a login page will be displayed.

Priority Level: High

Precondition: Users must give a unique username, valid email address and phone number.

3. Tracking system:

Real time tracking of buses. Display of bus location on map interface

Ability to search for buses by route or location

Priority Level: High

Precondition: The system has to access GPS data from buses and a database.

4. Reservation:

The people can reserve a seat for a particular bus. People can see the fare. People can choose their pickup bus stoppage. After reserve a seat, the payment option will be displayed.

Priority Level: Medium

Precondition: The user must have selected a bus route or specific bus journey.

5. Payment Options:

People can pay the fare through mobile banking system such as Baksh,

Nagad or rocket. If the user has insufficient balance the system will show that the seat cannot be booked because of insufficient balance. People will receive a receipt via email.

After payment the homepage will be displayed.

Priority Level: High

Precondition: Users have to valid mobile banking account; payment method should be supported by the system.

4.2 Non-functional Requirement

1. Performance

- ✦ Concurrent Users: The system shall handle up to 10,000 concurrent **users**.
- ✦ Response Time: The system shall respond to user queries within 2 seconds.
- ✦ Quantifiable: Handle up to 5000 concurrent users, respond within 5 seconds.
- ✦ Priority: High

2. Reliability

- ✦ Uptime: The system shall have 99.9% uptime.
- ✦ Recovery Time: The system shall recover from failures within 5 minutes.
- ✦ Quantifiable: Recover from failures within 2 minutes.
- ✦ Priority: High

3. Security

- ✦ Secure Communication: The system shall use HTTPS for all communications.
- ✦ Data Encryption: User data shall be encrypted both in transit and at rest.
Authentication: The system shall use strong authentication mechanisms.
- ✦ Quantifiable: encrypt user data in transit.
- ✦ Priority: High

4. Usability

- ✦ Intuitive Design: The system shall be intuitive and easy to use.
- ✦ Accessibility: The system shall be accessible to users with disabilities.
- ✦ Mobile-Friendly: The mobile application shall adhere to platform specific design guidelines.
- ✦ Priority: Medium

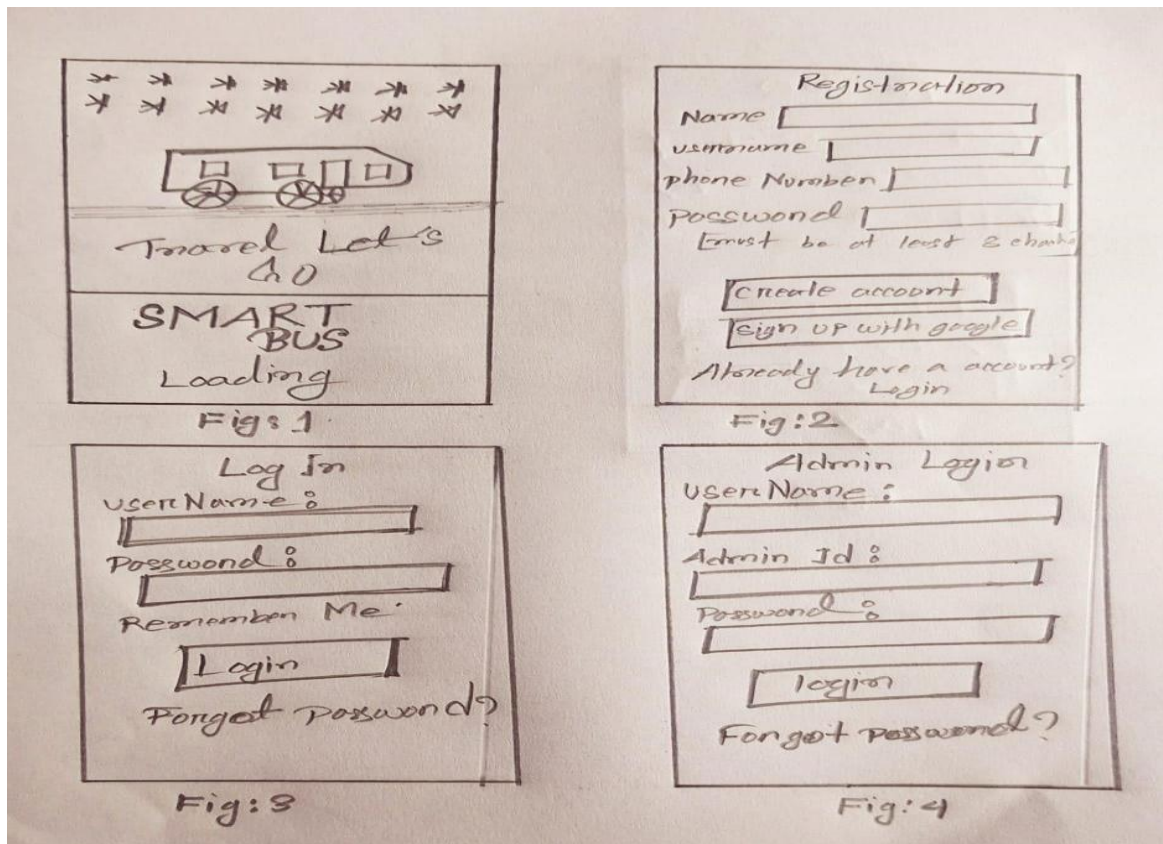
5. Maintainability

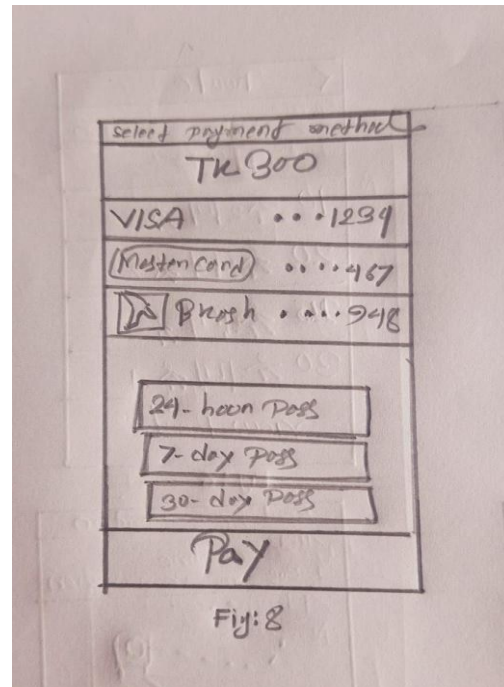
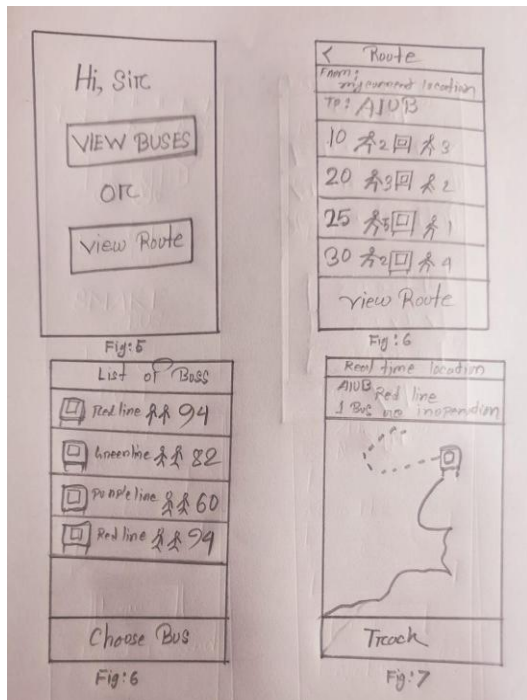
- † Code Quality: The system shall follow coding standards and best practices.
- † Documentation: The system shall include comprehensive documentation for developers and users.
- † Priority: Medium

6. Interoperability

- † Third-Party Integration: The system shall integrate with third-party mapping services (e.g., Google Maps).
- † Priority: Low

UI DESIGN





4.3 Project Development Constraints:

1. Technology Constraints:

Development Platform: The system shall be developed using specific technologies (JavaScript).

Database: The system shall use a specific database management system (MySQL). Priority: High

2. Resource Constraints:

Budget: The project shall adhere to a defined budget.

Timeframe: The project shall be completed within a specified timeframe (2 months).

Priority: High

3. Constraints:

Data Privacy: The system shall comply with data privacy laws General Data Protection Regulation (GDPR), Central Consumer Protection Authority (CCPA) Accessibility Standards: The system shall comply with accessibility standards (WCAG 2.1).

Priority: High

4. Operational Constraints:

GPS Accuracy: The system's real-time tracking depends on the accuracy of GPS data.

Network Connectivity: The system requires reliable internet connectivity for real-time updates.

Priority: Medium

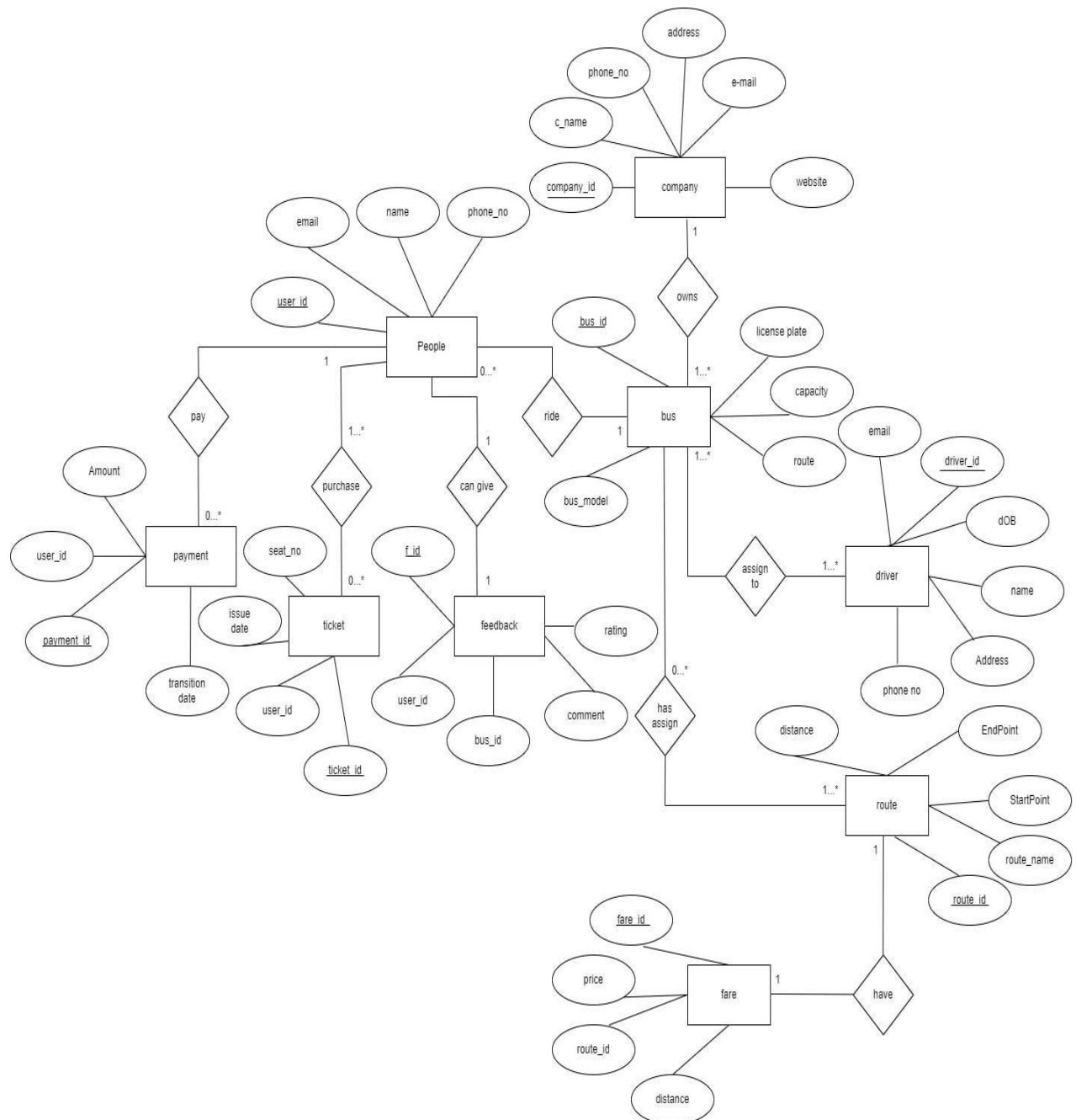
5. Deployment Constraints:

Hosting Environment: The system shall be hosted on a specified cloud platform (e.g., -).

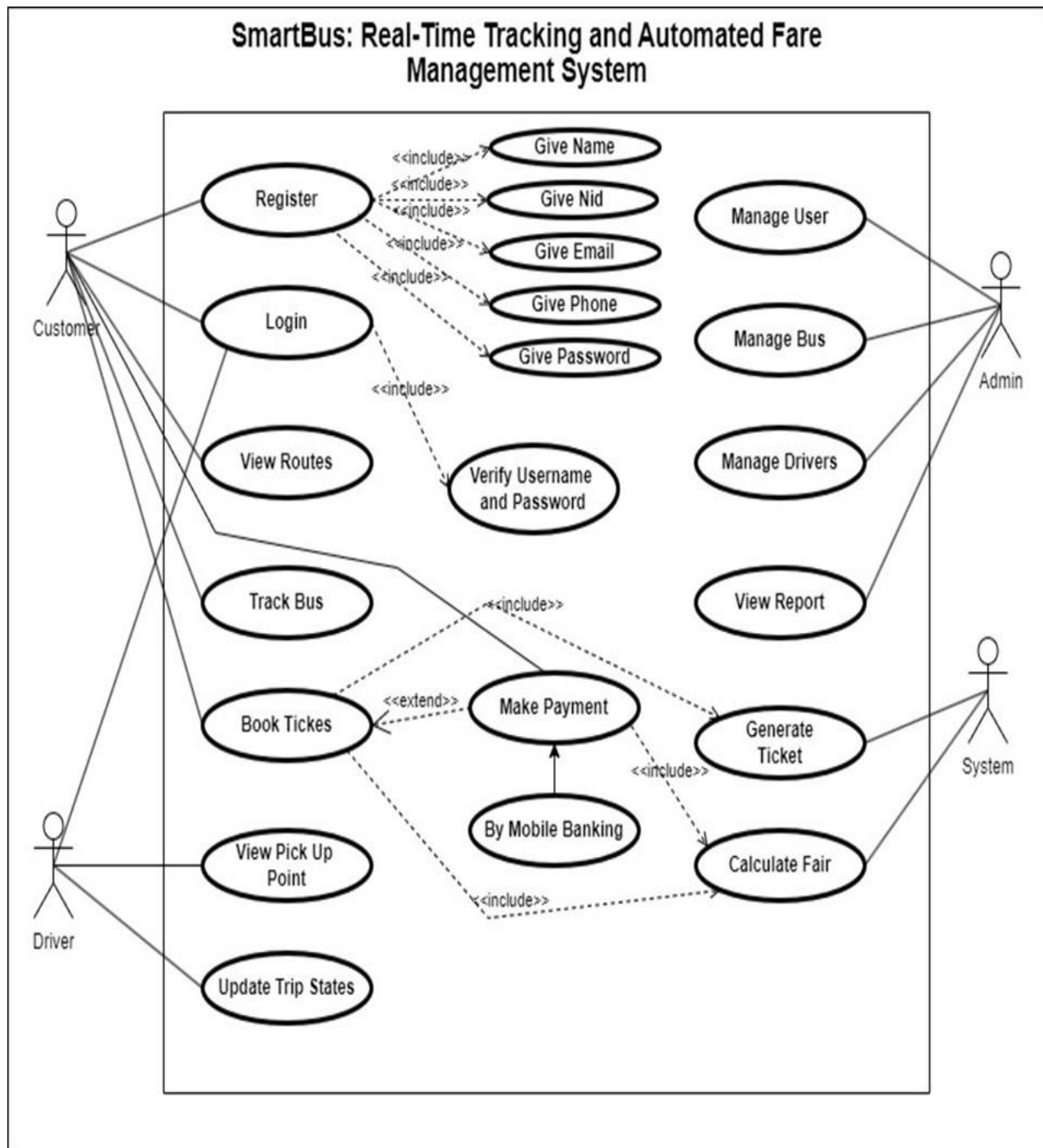
Deployment Schedule: The system shall follow a specific deployment schedule with minimal downtime.

Priority: Low

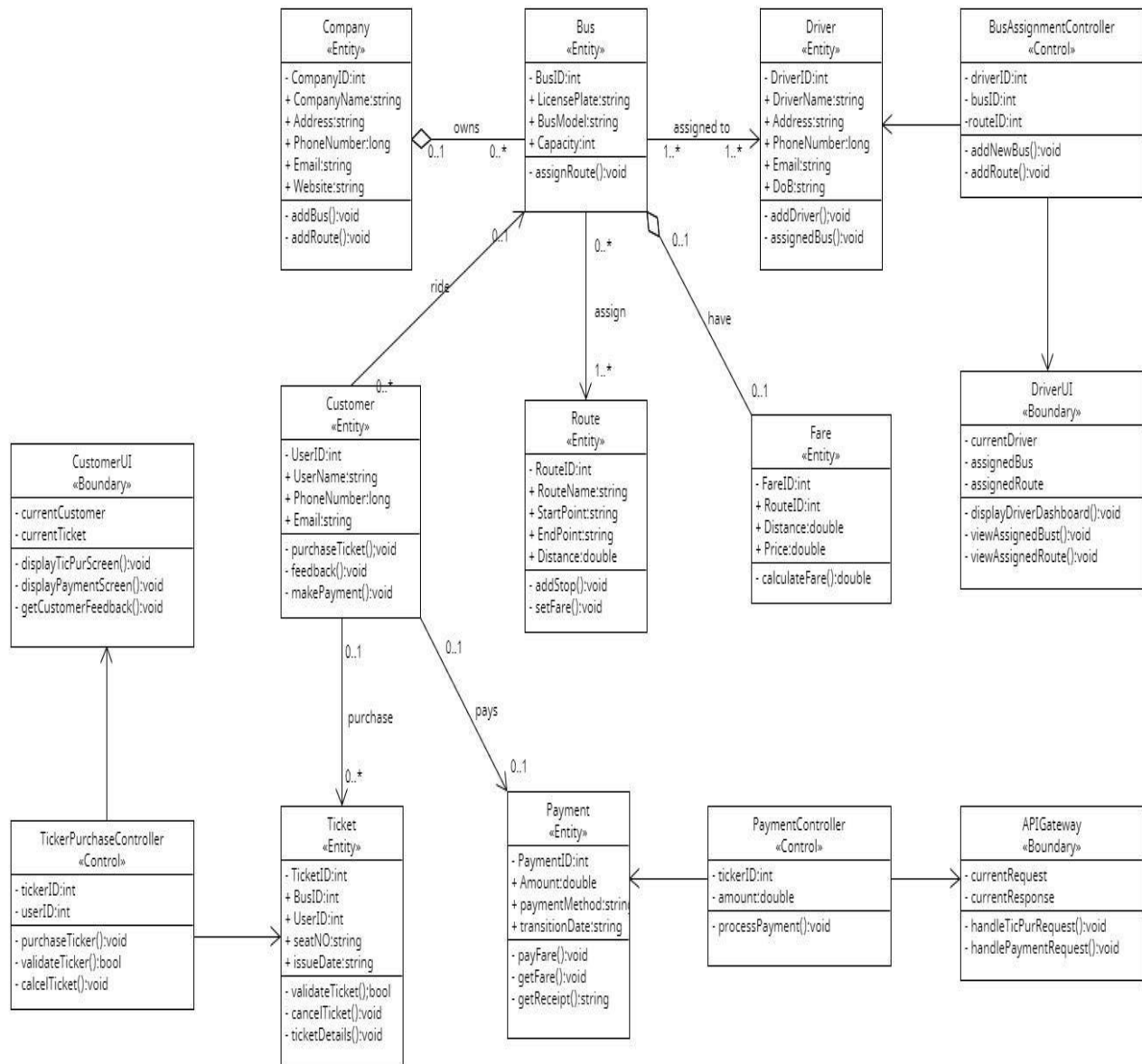
5.1 Er Diagram

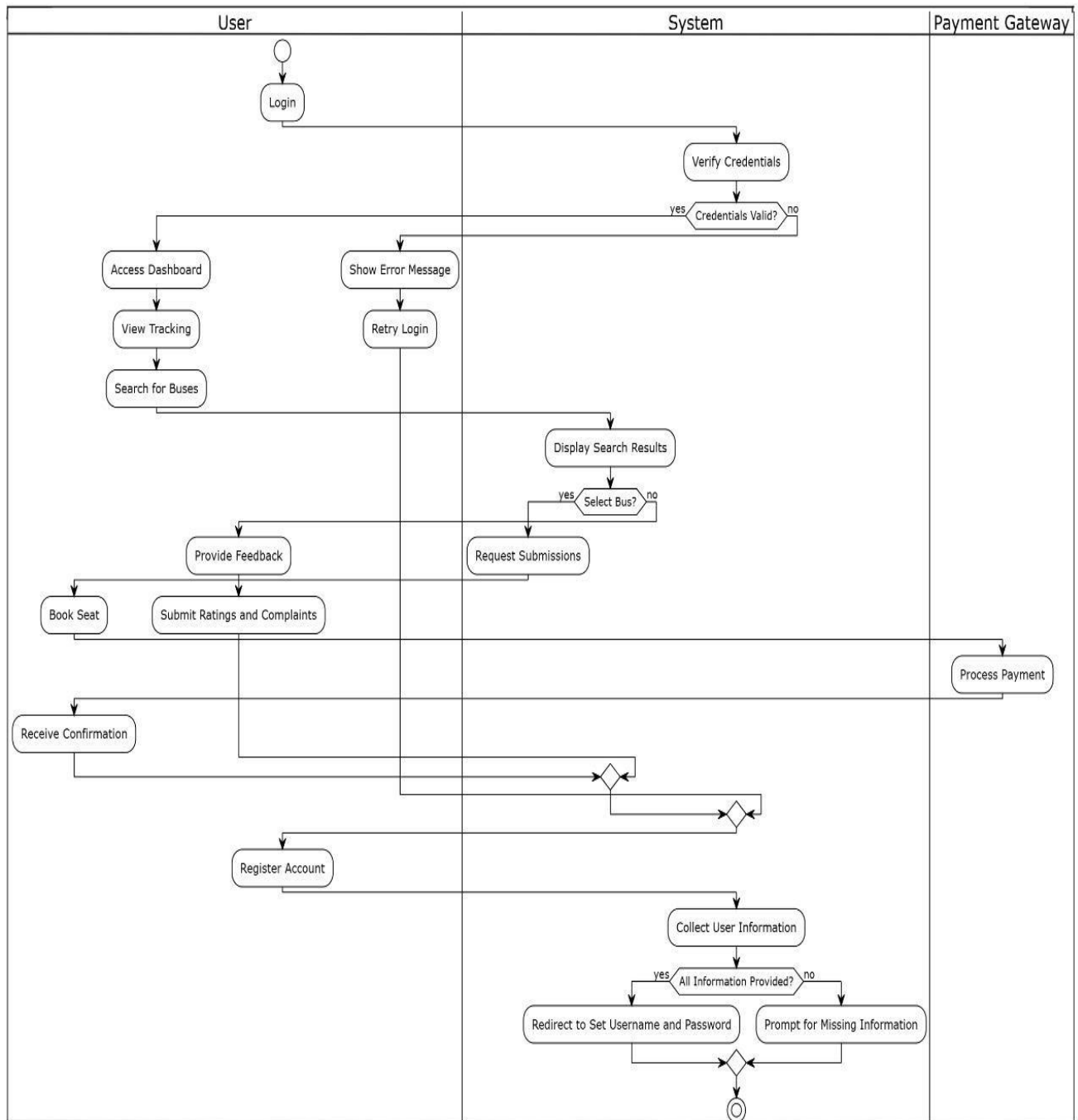


Use Case Diagram



Class Diagram



Activity Diagram:

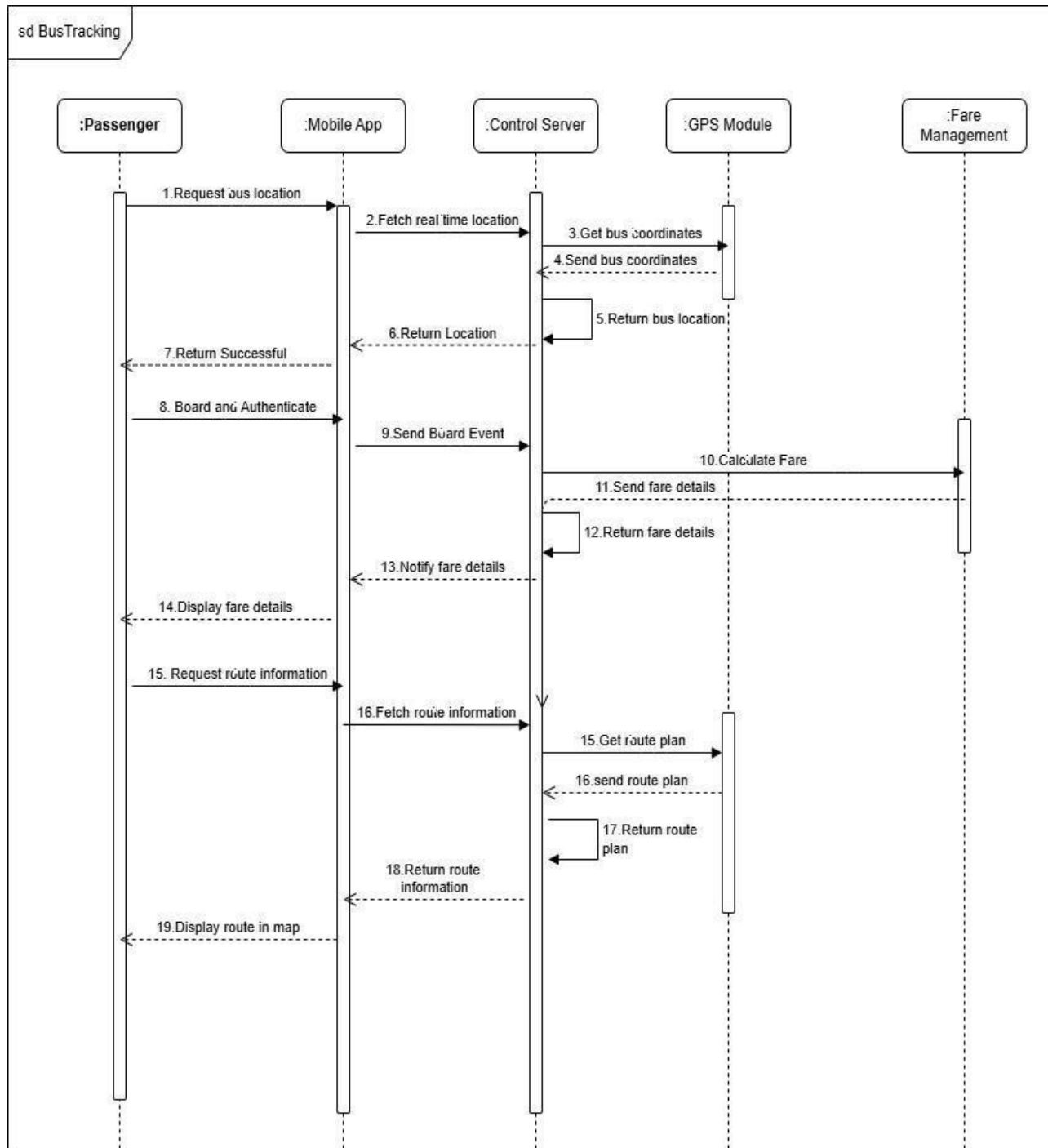
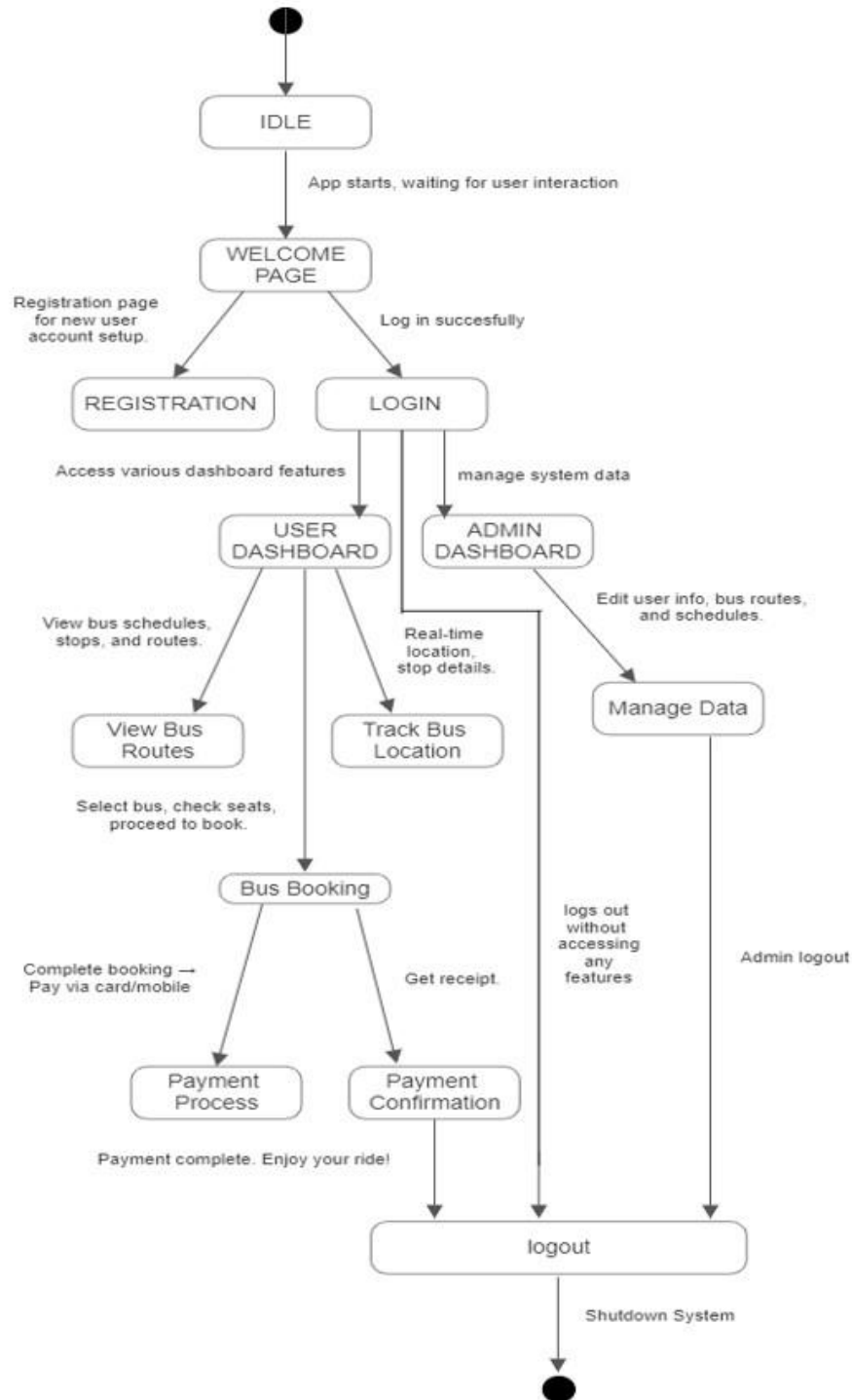
Sequence Diagram:

Chart diagram State



UI Design:

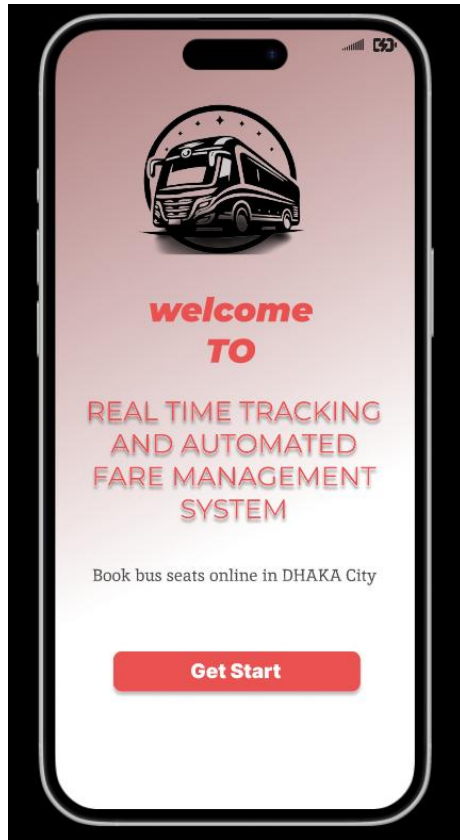


Fig:1

Fig:1

This is the welcome page of the application.



Fig:2

Fig:2

This is the registration page of the application. New users can register to the application through this page by providing full name, username, phone number and unique password.

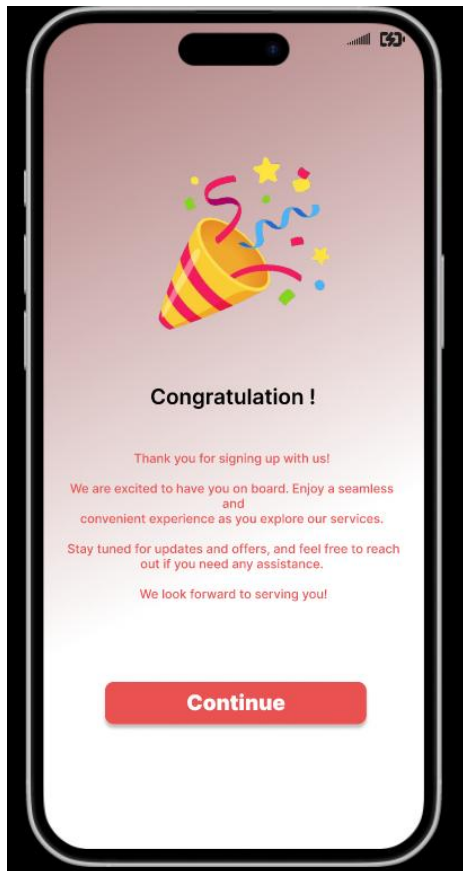


Fig:3

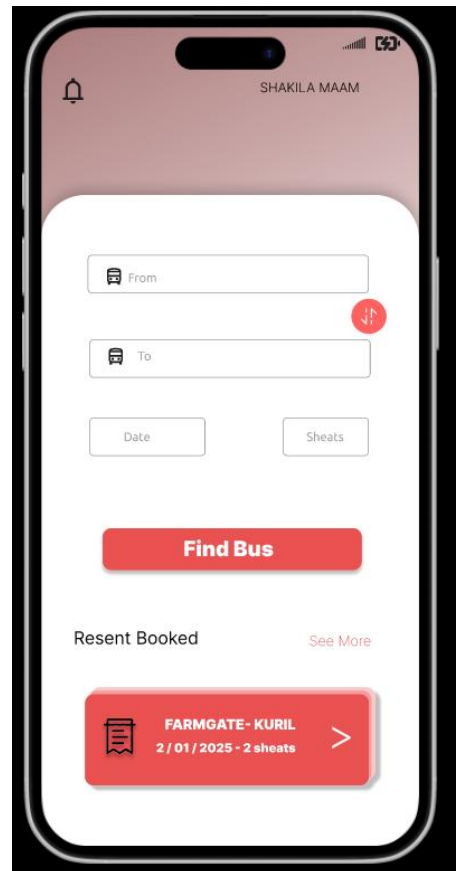


Fig:4

Fig:3

Thank you for coming to the new page by registering for real-time tracking and an automated fare management system.

Fig:4

Search for buses by specifying your departure and destination locations, along with the date and number of seats.

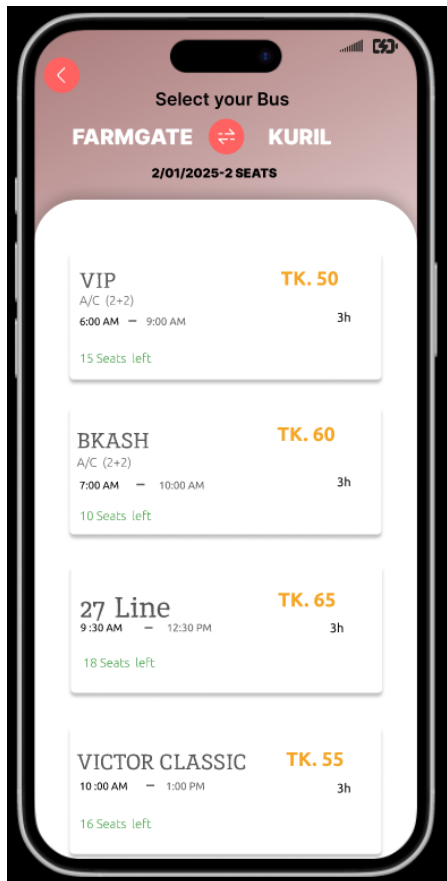


Fig:5

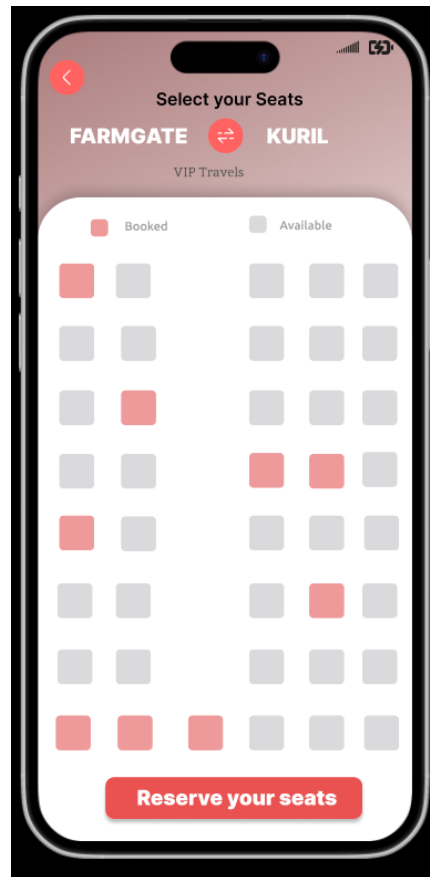


Fig:6

Fig:5

User can choose the bus for long distance. User can see the number of available seat. And also the bus fare. For these bus, user can book the seat. And pay the bus fare through various method.

Fig:6

The user can select their preferred seat from here to reach their destination. Booked seats will be marked in red, and they can select up to two seats at a time.

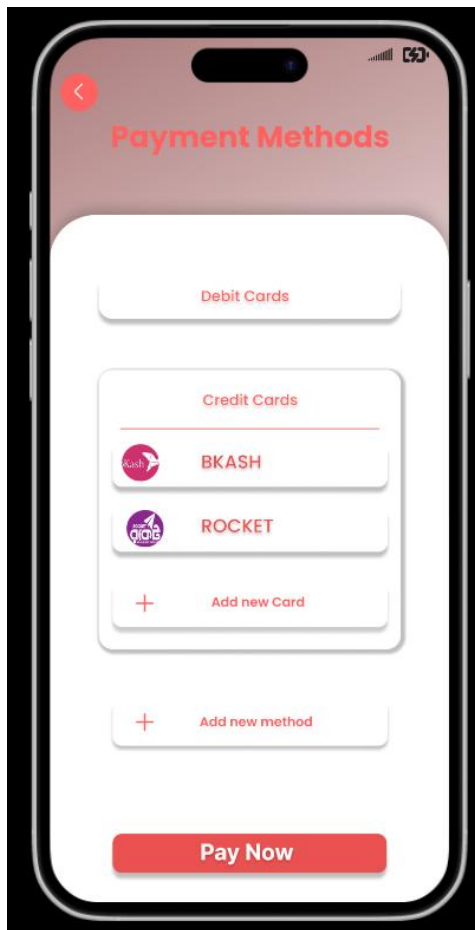


Fig:7



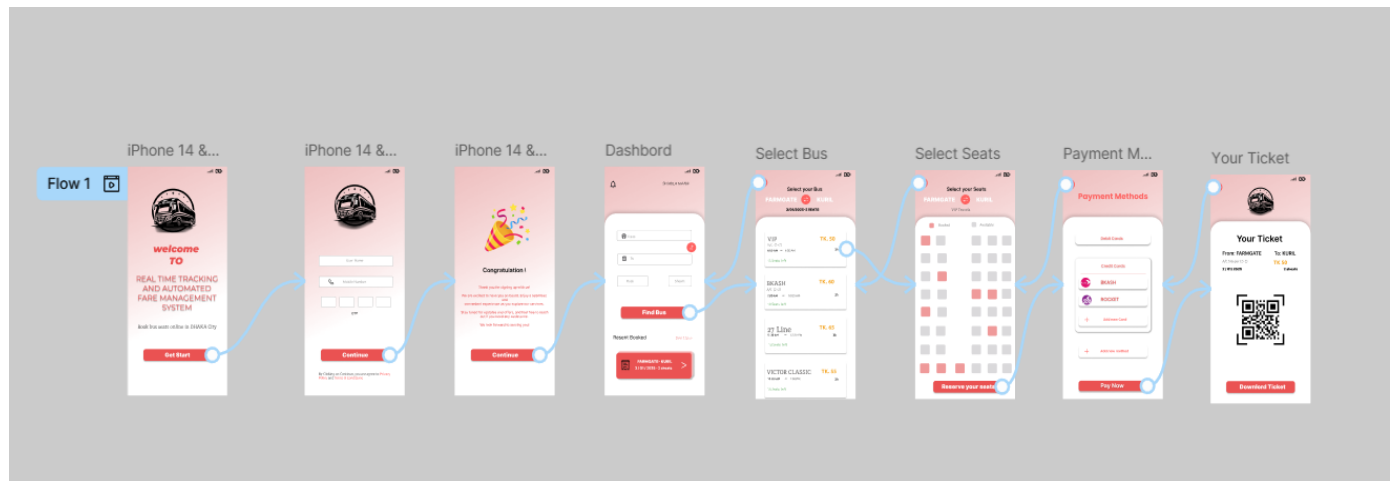
Fig:8

Fig:7

The user can make payments using debit cards, credit cards, and mobile banking systems such as bKash and Rocket. In this interface, the user can also add a new card and a new payment method. Once everything is completed correctly, the payment can be processed.

Fig:8

Once everything is completed correctly, the user will be able to collect their ticket. By clicking the download option, the ticket will be downloaded. The ticket can also be accessed by scanning it with a scanner.



The full prototyping Ui system

Here is the prototyping UI design link:

<https://www.figma.com/proto/ekqGeMN4cZ85SnRIPWEdx7/REAL-TIME-TRACKING-AND-AUTOMATED-FARE-MANAGEMENT-SYSTEM?node-id=1-2&p=f&t=9zsjWhb1VZkQuGEO-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A2>

Testing case

Project Name: Bus Tracking and Fair Management			Test Designed by: Masudur Rahman	
Test Case ID: SB 1.2			Test Designed date: 1/01/2025	
Test Priority (Low, Medium, High): Medium			Test Executed by: Masudur Rahman	
Module Name: Registration			Test Execution date: 10/01/2025	
Test Title: Verify user registration with valid input data				
Description: Registration process				
Precondition (If any): User should be on the registration page.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the application. 2. Click on the registration button. 3. Enter unique username, full name, valid phone number and unique password. 4. Click on the register button.	Username: Full Name: Masudur Rahman Phone no: 017*****70 Password: 101010	Users should register into the application.	User can't register in the application for invalid phone number	Fail
Post Condition: User data such as username, full name, phone no and password are added in the database.				

Project Name: Bus Tracking and Fair Management			Test Designed by: Masudur Rahman	
Test Case ID: SB 1.4			Test Designed date: 1/01/2025	
Test Priority (Low, Medium, High): High			Test Executed by: Masudur Rahman	
Module Name: Login			Test Execution date: 10/01/2025	
Test Title: Verify login with valid username and password				
Description: Login process				
Precondition (If any): User should have valid username and password.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the application. 2. Enter the username and password. 3. Click on the login button.	Username: Masudur Rahman Password: 101010	Users should be logged in the application.	User can't login to the application for incorrect password	Fail
Post Condition: User data are checked with the database and successfully login to the application.				

Project Name: Bus Tracking and Fair Management			Test Designed by: Masudur Radman	
Test Case ID: SB 1.6			Test Designed date: 1/01/2025	
Test Priority (Low, Medium, High): High			Test Executed by: Masudur Rahman	
Module Name: Admin Login			Test Execution date: 10/01/2025	
Test Title: Verify admin login with valid data				
Description: Admin login process				
Precondition (If any): Admin should have verified data				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to the application. 2. Click on the admin login button. 3. Enter admin username, id, password. 4. Click on the login button.	Username: Masudur Rahman Password: 101010	User should login to the application.	As expected,	Pass
Post Condition: User data are checked with the database. As it is not matched, admin will be redirected to the home page.				

Project Name: Bus Tracking and Fair Management			Test Designed by: K.M.M. SAMIN	
Test Case ID: SB 1.8			Test Designed date: 1/01/2025	
Test Priority (Low, Medium, High): Medium			Test Executed by: K.M.M. SAMIN	
Module Name: Bus location			Test Execution date: 10/01/2025	
Test Title: Locate the bus location on the map				
Description: GPS integration for real time bus location				
Precondition (If any): Bus name should be valid				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/ Fail)
1. Go to the application. 2. Login to the application. 3. Go to the location on the bus page. 4. Enter the bus name. 5. Click on the search button.	Bus name: VIP	Bus location should be shown on the map.	The bus location is not shown in the application as the bus is not registered in the application.	Fail
Post Condition: Bus location should be real time.				

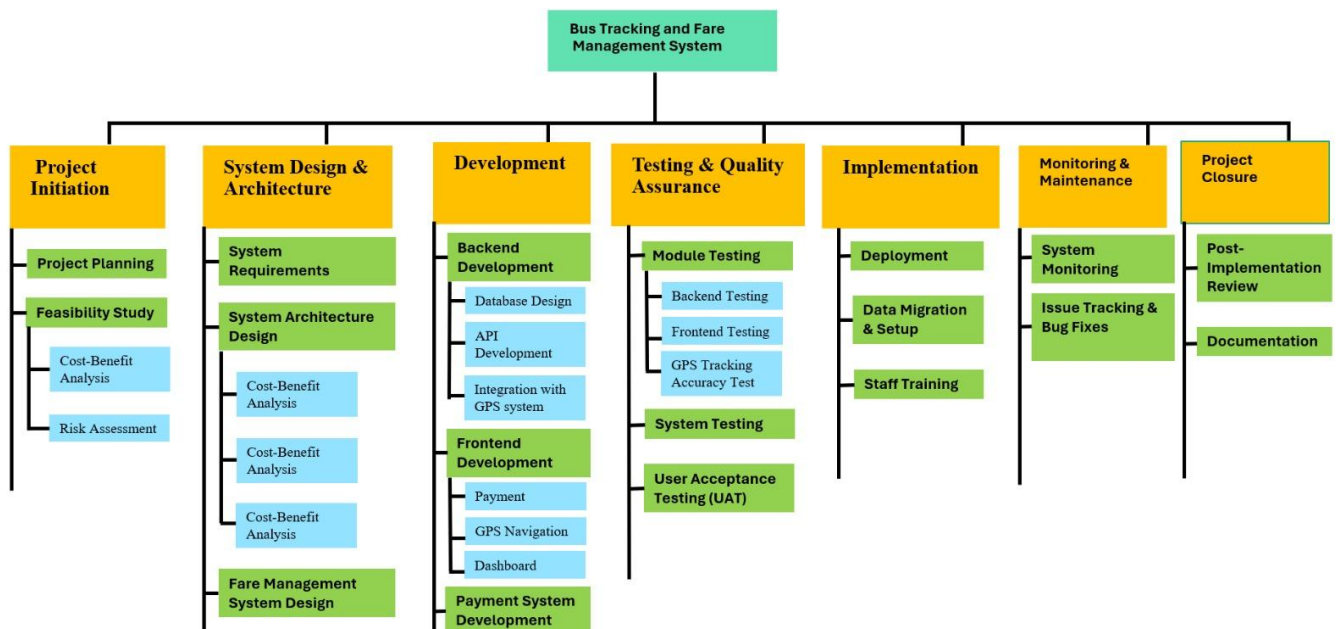
Project Name: Bus Tracking and Fair Management			Test Designed by: K.M.M. SAMIN	
Test Case ID: SB 2.0			Test Designed date: 1/01/2025	
Test Priority (Low, Medium, High): Medium			Test Executed by: K.M. SAMIN	
Module Name: add bus			Test Execution date: 10/01/2025	
Test Title: add or remove bus and route in the application.				
Description: New bus and route will be included and modify the remaining.				
Precondition (If any): User must be admin.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/ Fail)
1. Go to the application. 2. Click on the admin login button. 3. Enter admin username, id, password. 4. Click on the login button. 5. Go to the modify page. 6. Click on the New button. 7. Enter the bus name, bus id, bus route. 8. Click the add button.	Bus name: BKASH Bus ID: 101 Route: Kuril to Farmgate	It should be added in the application.	Bus is not added in the application because the bus has not a unique id.	Fail
Post Condition: New bus data should be synced in the database.				

Project Name: Bus Tracking and Fair Management			Test Designed by: K.M.M. SAMIN	
Test Case ID: SB 2.2			Test Designed date: 1/01/2025	
Test Priority (Low, Medium, High): High			Test Executed by: K.M.M. SAMIN	
Module Name: Route dashboard			Test Execution date: 10/01/2025	
Test Title: Bus information				
Description: Get the bus info according to the start and destination				
Precondition (If any): User should input valid location.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/ Fail)
1. Go to the application. 2. Login to the application. 3. Go to the route dashboard page. 4. Enter the start location and destination. 5. Click on the search icon.	Start location: farmgate Destination: Kuril flyover	Buses name and fare should be shown.	Start location or destination is not valid.	Fail
Post Condition: The information will be saved in the database.				

Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) for the Bus Tracking and Fare Management System provides a comprehensive framework for project execution and management. The **Project Initiation** phase lays the groundwork with project planning, feasibility studies, cost-benefit analysis, and risk assessment. The **System Design & Architecture** phase focuses on defining system requirements, developing the architecture with financial viability evaluations, and designing the automated fare management system. The **Development** phase involves backend tasks such as database design, API creation, and GPS integration, alongside frontend efforts including payment system development, GPS navigation, and dashboard design. **Testing & Quality Assurance** ensures system functionality through module testing (backend, frontend, and GPS accuracy), system testing, and user acceptance testing (UAT). The **Implementation** phase includes system deployment, data migration, setup, and staff training to prepare for live operations. In **Monitoring & Maintenance**, system performance is continuously observed, and issues are tracked and resolved promptly. Finally, the **Project Closure** phase formalizes the project's conclusion with a post-implementation review and detailed documentation. This structured WBS ensures the successful design, development, deployment, and maintenance of a reliable real-time tracking and fare management system.

WORK BREAKDOWN STRUCTURE (WBS)



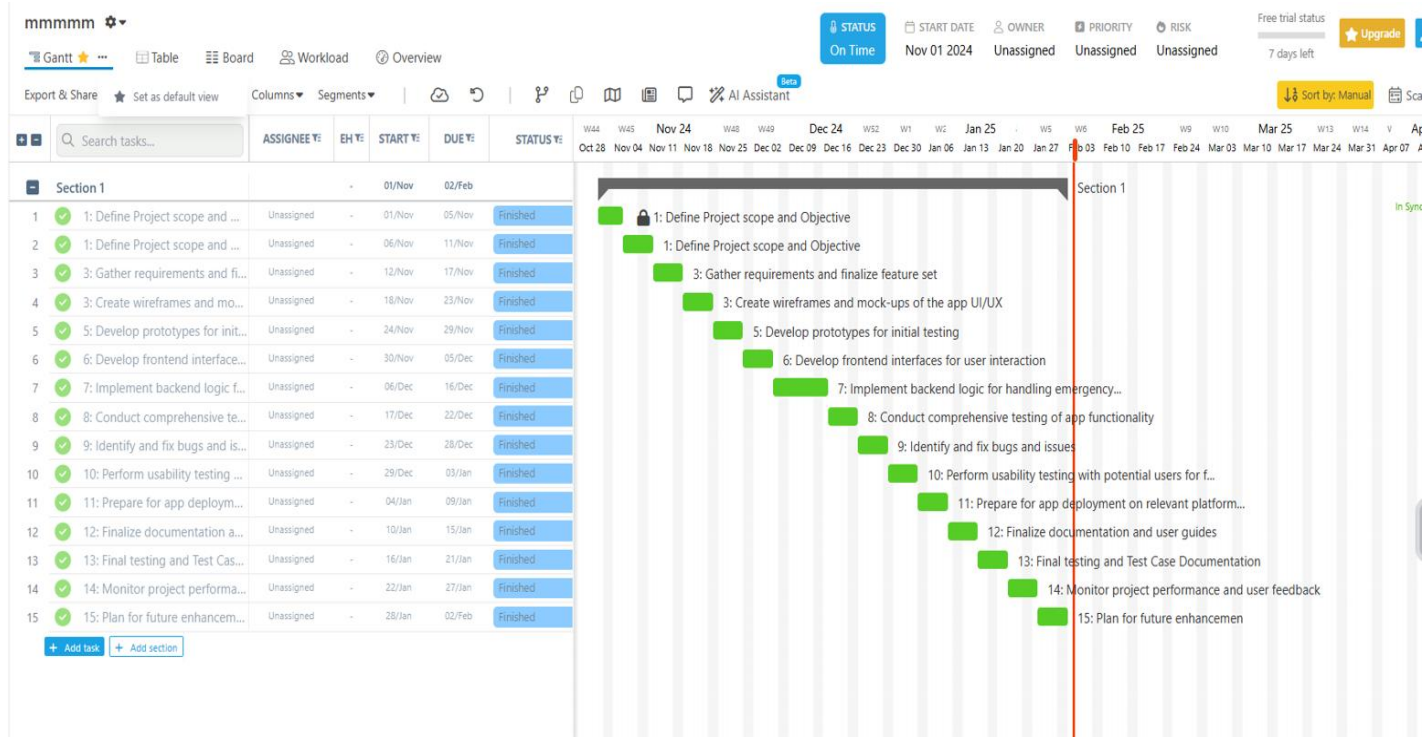
Risk Analysis for Real-Time Tracking and Automated Fare Management System

The Risk Analysis for the Real-Time Tracking and Automated Fare Management System identifies critical challenges that could affect the project's success and provides strategies to mitigate them. Key risks include GPS tracking system failures, which could lead to inaccurate bus location data, mitigated by redundant GPS modules and failover systems. Integration issues between APIs and subsystems may disrupt data exchange, addressed through rigorous integration testing. Data migration errors during implementation pose a high risk of data loss or corruption, requiring pilot migrations and backups. Cybersecurity threats to the payment system, which could result in data breaches or financial losses, will be countered with robust encryption and firewalls. User resistance to new technology might delay adoption, managed through comprehensive training and user-friendly system designs. Hardware failures in buses or monitoring stations can disrupt operations, but regular maintenance and spare part inventories will minimize downtime. Financial overbudgeting due to unforeseen complexities is mitigated by detailed cost estimation and monitoring. Inaccurate fare calculations could lead to user dissatisfaction, necessitating extensive algorithm testing. Insufficient system performance during peak hours, threatening reliability, will be addressed through architecture optimization and load balancing. Regulatory compliance issues, which could result in delays or penalties, will be handled by engaging legal experts during system design. Assigning risk ownership and implementing these strategies ensures the project's resilience and successful execution.

ID	Description of Risk	Impact	Risk Response	Risk Level	Risk Owner	Notes
1	GPS tracking system failure	Inaccurate real-time bus location data	Implement redundant GPS modules and failover systems	High	System Architect	Requires regular maintenance and testing of GPS modules
2	Integration issues between APIs and subsystems	Data exchange failures	Perform rigorous integration testing	Medium	Development Team	APIs should be thoroughly documented and tested before deployment.
3	Data migration errors during implementation	Loss or corruption of critical system data	Conduct pilot migrations and create backups	High	Implementation Manager	Ensure proper validation of data post-migration.
4	Cybersecurity threats to the payment system	Data breaches, financial losses	Implement robust encryption and firewalls	High	Security Specialist	Regularly update software to address vulnerabilities.
5	User resistance to new technology	Delayed adoption and inefficient operations	Provide comprehensive user training	Medium	Training Coordinator	Incorporate user feedback to simplify system interfaces.
6	Hardware failures in buses or monitoring stations	Downtime and operational disruptions	Maintain an inventory of spare parts	Low	Maintenance Team	Schedule routine hardware checks and preventive maintenance.
7	Overbudget due to unforeseen development complexities	Financial strain on the project	Conduct detailed cost estimation and include buffer	Medium	Project Manager	Monitor costs and reallocate resources as needed.
8	Inaccurate fare calculations in the automated system	User dissatisfaction and revenue loss	Test fare calculation algorithms extensively	High	Testing Team	Perform real-world scenario testing before deployment.

9	Insufficient system performance during peak hours	Reduced reliability and user trust	Optimize system architecture and load balancing	High	System Architect	Conduct stress testing to ensure performance under heavy load.
10	Regulatory compliance issues	Project delays or legal penalties	Engage legal experts during system design	Low	Compliance Officer	Ensure all legal and regulatory requirements are met.

GANTT-CHART



Activity Key:

A. Project Planning	J. Module Design 3
B. Requirements Gathering	K. Code Module 1
C. Feasibility Study	L. Code Module 2
D. Create System Architecture	M. Code Module 3
E. UI Design	N. Integration
F. Schema Design	O. Integration Testing
G. Frontend Development	P. System Testing
P. Module Design 1	Q. Post-launch Monitoring
Q. Module Design 2	R. Documentation and Training

Effort Estimation

1. Requirement Analysis & Specification (1-2 Weeks)

- **Tasks:**
 - Gather functional and non-functional requirements.
 - Define the system architecture (frontend, backend, and data integration).
 - Identify stakeholders (passengers, administrators).
 - Prepare the project plan and technical documentation.
- **Effort:**
 - **Team Members Involved:** Business Analyst, Project Manager, Tech Lead
 - **Estimated Time:** 1-2 weeks
 - **Total Effort:** 40-80 person-hours

2. System Design (2-3 Weeks)

- **Tasks:**
 - Define the architecture: Client-side application (Mobile/Web), Server (Backend), and GPS device integration.
 - Design database schema for storing real-time data.
 - Create API design for communication between client, backend, and GPS tracking services.
 - Design UI/UX for passenger and admin interface.
- **Effort:**
 - **Team Members Involved:** Solution Architect, UX/UI Designer, Database Architect, Tech Lead
 - **Estimated Time:** 2-3 weeks
 - **Total Effort:** 80-120 person-hours

3. Development (6-10 Weeks)

3.1 Backend Development

- **Tasks:**
 - Develop API endpoints for bus tracking.
 - Implement real-time data processing using GPS tracking data.
 - Integration with third-party GPS services (or building custom GPS solution).
 - Handle data persistence, authentication, and user management for admin and passenger portals.
- **Effort:**
 - **Team Members Involved:** Backend Developer, Tech Lead
 - **Estimated Time:** 4-5 weeks

- **Total Effort:** 160-200 person-hours

3.2 Frontend Development

- **Tasks:**
 - Develop web/mobile app for passengers to view real-time bus locations.
 - Build an admin dashboard for monitoring fleet status and generating reports.
 - Display bus routes, stops, and ETAs on an interactive map.
- **Effort:**
 - **Team Members Involved:** Frontend Developer, UI/UX Designer
 - **Estimated Time:** 3-4 weeks
 - **Total Effort:** 120-160 person-hours

3.3 Database Development

- **Tasks:**
 - Implement and optimize database queries for storing bus routes, stops, real-time locations, and user information.
 - Ensure scalability and performance for handling real-time data streams.
- **Effort:**
 - **Team Members Involved:** Database Developer
 - **Estimated Time:** 2-3 weeks
 - **Total Effort:** 40-60 person-hours

4. Testing & Quality Assurance (3-4 Weeks)

- **Tasks:**
 - Unit testing for individual modules (frontend, backend).
 - Integration testing between APIs, frontend, and GPS devices.
 - Performance testing to ensure real-time tracking functionality.
 - User acceptance testing (UAT) with passengers and administrators.
- **Effort:**
 - **Team Members Involved:** QA Engineer, Tech Lead
 - **Estimated Time:** 3-4 weeks
 - **Total Effort:** 120-160 person-hours

5. Deployment & Setup (1-2 Weeks)

- **Tasks:**
 - Set up production environment (cloud servers, databases).
 - Deploy the backend, frontend, and any third-party integrations.
 - Train administrators on using the system.
- **Effort:**
 - **Team Members Involved:** DevOps Engineer, Tech Lead, Project Manager
 - **Estimated Time:** 1-2 weeks
 - **Total Effort:** 40-80 person-hours

6. Maintenance & Support (Ongoing)

- **Tasks:**
 - Bug fixing, updates, and ongoing feature improvements.
 - Regular performance monitoring.
 - Provide customer support and handle feedback.
- **Effort:** This is an ongoing process post-launch and depends on user adoption, changes, and system needs.
 - **Team Members Involved:** Support Team, Development Team
 - **Effort:** 20-40 person/hours (post-launch)

Total Effort Estimation

Phase	Estimated Effort (Person-Hours)
Requirement Analysis & Specification	40-80
System Design	80-120
Backend Development	160-200
Frontend Development	120-160
Database Development	40-60
Testing & Quality Assurance	120-160
Deployment & Setup	40-80
Total Development Effort	600-860 hours

As the project is strongly coupled to hardware environment. So, it is an embedded project. So, the embedded values of the “**Bus tracking and fair management system**” is

SLOC = 80,000.

P = 1.20, T = 0.32, Coefficient_{<Effort Factor>} = 3.6

$$\begin{aligned}
 \text{Effort} = \text{PM} &= \text{Coefficient}_{\text{Effort Factor}} * (\text{SLOC}/1000)^P \\
 &= 3.6 * (80000/1000)^{1.20} \\
 &= 691.85 \\
 &\approx 692
 \end{aligned}$$

$$\text{Development time} = \text{DM} = 2.50 * (\text{PM})^T$$

$$= 20.267$$
$$= 2.50 \cdot (692)^{0.32}$$
$$\approx 21$$

Required number of people = $ST = PM/DM$

$$= 692/21$$
$$\approx 33$$

“THANKS”